Adding and Subtracting Rational Expressions

A ratio compares two numbers and can be written with words, with a colon, or as a fraction. For example, the ratio "3 to 5" can be written as 3:5 or $\frac{3}{5}$.

A **rational expression** is a ratio that compares two polynomials. Rational expressions are also written as fractions. Both the numerator and the denominator of a rational expression must be polynomials. Rational expressions can't have fractional exponents, negative exponents, or radicals.

The rules that apply to working with fractions apply to rational expressions. When you add or subtract fractions with like denominators, add or subtract the numerators and keep the denominator the same. You can add rational expressions the same way.

Example 1

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Add: $\frac{4m^2}{n^3} + \frac{m^2-3m}{n^3}$

Step 1

The denominators are the same. Add the numerators.

$$\frac{4m^2}{n^3} + \frac{m^2-3m}{n^3} = \frac{4m^2+m^2-3m}{n^3}$$

Step 2

Simplify the numerator.

$$\frac{4m^2+m^2-3m}{n^3} = \frac{5m^2-3m}{n^3}$$

When adding or subtracting rational expressions with unlike denominators, you need to find the lowest common denominator. The lowest common denominator must have the least common multiple of the number coefficients. It must also have each variable raised to the highest power that occurs in either denominator.

Example 2

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Subtract: $\frac{2}{3xy} - \frac{x+3}{5x^2y}$

Step 1

Find the lowest common denominator.

The least common multiple of 3 and 5 is 15.

The highest power of x is x^2.

The highest power of y is y.

The lowest common denominator is $15x^2y$.

Step 2

Rewrite both expressions using the lowest common denominator.

$$\frac{2}{3xy} \bullet \frac{5x}{5x} = \frac{10x}{15x^2y}$$

$$\frac{x+3}{5x^2y} \bullet \frac{3}{3} = \frac{3(x+3)}{15x^2y} = \frac{3x+9}{15x^2y}$$

Step 3

Subtract the expressions.

$$\frac{10x}{15x^2y} - \frac{3x+9}{15x^2y}$$

$$= \frac{10x-(3x+9)}{15x^2y}$$

$$= \frac{10x-3x-9}{15x^2y}$$

$$= \frac{7x-9}{15x^2y}$$

Some rational expressions have a binomial in the denominator. To add these rational expressions, simply multiply the binomials to find the common denominator.

Example 3

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Add: $\frac{7x}{x+4} + \frac{8}{x-3}$

Step 1

Find a common denominator by multiplying the denominators.

The lowest common denominator is $(x + 4)(x - 3)$.

Step 2

Rewrite the expressions using the common denominator.

$$\frac{7x}{x+4} \bullet \frac{x-3}{x-3} = \frac{7x(x-3)}{(x+4)(x-3)} = \frac{7x^2-21x}{(x+4)(x-3)}$$

$$\frac{8}{x-3} \bullet \frac{x+4}{x+4} = \frac{8(x+4)}{(x+4)(x-3)} = \frac{8x+32}{(x+4)(x-3)}$$

Step 3

Add and simplify. Leave the denominator in factored form.

$$\frac{7x^2-21x}{(x+4)(x-3)} + \frac{8x+32}{(x+4)(x-3)}$$

$$= \frac{7x^2-21x+8x+32}{(x+4)(x-3)}$$

$$= \frac{7x^2-13x+32}{(x+4)(x-3)}$$

Key Term

rational expression

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